Housing First or Treatment First?

A stylized model, and the quantitative model behind it

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1 A stylized model

1.1 Three questions

In the data, drug use is concentrated among low income households, most of the homeless do not carry a substance use disorder (SUD), and housing vouchers reach only about a quarter of the households eligible for them. The quantitative model is built to match facts of that kind. It has six state variables and fifty four parameters, which is what the welfare and fiscal questions require, but it also makes it hard to see which pieces of the environment do the heavy lifting. This note strips the environment down to help understand the main drivers in the model, but where my three main questions are still well posed.

Households are perfectly forward looking, here and in the quantitative model. They face the true hazard of acquiring a SUD, they know what carrying one would cost them, and they make no mistakes. Nothing below rests on misperception, on myopia, or on a taste that changes once the household has used.

1. **Who uses drugs?** In the data, use is concentrated at the bottom of the income distribution. Drugs here are a *luxury* good bought at a price: the Stone-Geary form in equation (1) below means the share of income spent on them rises with income, so left to itself the environment concentrates use at the *top*. Which feature of the environment reverses the concentration from the top to the bottom?
2. **How does housing assistance shape the drug use decision?** A voucher pulls drug use in two opposite directions. It raises the value of staying clean, because a housed future is worth more than a homeless one and using puts that future at risk: call this the value of life force, and it lowers use. In contrast, it cushions the consequence of acquiring a SUD, because a household that acquires one is housed anyway: call this the moral hazard force, and it raises use. Both are first order, so theory does not sign the effect of housing assistance on drug use, and the question is quantitative.
3. **When is Treatment First better than Housing First?** In a model without externalities, never. Housing First always raises welfare when agents are forward looking, as they are here. A SUD creates a genuine tension, since the same voucher that insures a household also makes the risk it insures more attractive, but that tension does not raise welfare, because Treatment First only removes an option that a household can take under Housing First as well. However, if a SUD and homelessness generate externalities, such as affecting the earning potential of the children of parents with a SUD, then Treatment First may be preferred: children of a parent with a SUD are themselves two to four times as likely to develop one (McGovern et al., 2023), and moving a child to a lower poverty neighbourhood before age thirteen raises adult earnings by 31 percent (Chetty et al., 2016). The model's work here is to show why Housing First raises welfare whenever there are no externalities, and then to qualitatively price the externality that overturns it.

The model is qualitative throughout: the calibration is rough by design, there is no validation exercise, and every component that obscures the mechanisms has been removed and the model refitted without it. What remains are the main drivers of the results. The welfare evaluations below are for illustration only; the full quantitative model is what settles the main question, which is whether Treatment First raises welfare relative to Housing First in a model that fits the data.

1.2 The environment

1.2.1 Households and preferences

There are two periods and a continuum of households. A period stands for thirty years of adult life, so the two of them span ages twenty five to eighty five. Parameters with a natural annual counterpart are

compounded to that length: the discount factor is β per period, the gross interest rate is $R = 1/\beta$ so that $\beta R = 1$, and ϕ_0 is survival over thirty years. Notation follows the quantitative model throughout.

A household is characterised by its productivity y, fixed over its life and lognormally distributed across households, $\log y \sim \mathcal{N}(-\frac{1}{2}\sigma_y^2, \sigma_y^2)$, so mean income is one. Households differ in y, in age $j \in \{1, 2\}$, in employment status $e \in \{0, 1\}$, in voucher receipt $\eta \in \{0, 1\}$ and in SUD status $d \in \{0, 1\}$, all discussed below, and in nothing else.

Flow utility depends on non durable consumption c, housing services h and drug consumption v,

$$u(c, h, v) \;=\; (1 - \gamma - \theta) \log c \;+\; \gamma \log h \;+\; \theta\big[\log(b + v) - \log b\big] \;+\; \aleph. \tag{1}$$

The three weights sum to one, so γ is the housing budget share. Drugs enter only in the first period, since that is the only period in which v is chosen, and housing is chosen each period from a three point grid $\{h_1, h_2, h_3\}$. The pair (θ, b) governs the drug term: the marginal utility of the vth unit is $\theta/(b + v)$, so θ mostly sets how much a user uses and b whether anyone uses at all. The Stone-Geary form keeps the first unit's marginal utility finite, which is what generates non users at all, because under plain $\theta \log v$ every household uses whatever the consequences. Finally $\aleph$ is the flow value of being alive, the utility of surviving into a period over and above what is consumed in it, which gives mortality a cost that does not vanish when consumption is low. Households save in one risk free asset and cannot borrow.

1.2.2 Timing

Period 1. Every household is clean. Employment $e \in \{0, 1\}$ is drawn with the unemployment probability u_0, and a household that is income eligible draws a voucher with probability π_V; these are the same two lotteries it faces again in period 2. It then chooses consumption, housing, drug use $v \geq 0$, which for simplicity is the only period in which it chooses whether to use, and saving, paying market rent p_r, which is normalised to one, on whatever unit it picks. It starts with no assets. Using drugs generates the risk of a SUD, increasing in v,

$$\pi(v) \;=\; 1 - e^{-\lambda v}. \tag{2}$$

Because π is increasing in v, the household internalises that using more is riskier, so the intensive and extensive margins are not separate decisions.

The transition. The SUD $d' \in \{0, 1\}$ is realised with probability $\pi(v)$ for households that used and with probability zero for the rest, and the household survives into period 2 with probability

$$\phi(d') \;=\; \phi_0\,(1 - \kappa d'), \tag{3}$$

so κ is the proportional reduction in survival caused by the SUD. The mortality penalty applies here, and here only.

Period 2. Terminal, after which the household dies with certainty. A household without the SUD is unemployed with probability u_0 and one with the SUD with the higher probability u_1; if unemployed it receives income support. The voucher is drawn again, subject to the policy rule. The household chooses consumption and housing, there is no drug use decision, and assets are run down with nothing bequeathed.

Mortality, and where the SUD is realised. Mortality is a tax on a continuation value, so the SUD can only cost anything if a future remains when it is carried. That does not require a third period. It requires the SUD to be drawn at a *transition* rather than inside the terminal period, and here it is drawn on the way into period 2, with survival over that transition depending on it, so acquiring it puts the whole of period 2 at risk. Realise it inside the last period instead and death costs nothing because nothing follows it, and the mechanism the model rests on falls silent.

1.2.3 Employment, income support and housing

The gap $u_1 - u_0$ is the entire resource cost of the SUD. Income in a period, given permanent productivity y and employment status e, is

$$y^g(y,e) \;=\; \begin{cases} y & \text{if } e=1, \\ \psi \cdot \min\{y, \bar{y}\} & \text{if } e=0, \end{cases} \tag{4}$$

with ψ the replacement rate and $\bar{y}$ a cap on the earnings base. There is no other income risk. The level of ψ is what links the SUD to homelessness: for the poorest households half of income can fall short of the rent on the smallest unit, so an unemployed household there cannot house itself out of current income.

Each period the household chooses

$$h_k \in \{h_1,\ h_2,\ h_3\}, \qquad h_1 < h_2 < h_3, \tag{5}$$

homelessness (h_1), a small unit (h_2) and a standard unit (h_3). Homelessness is free and delivers less housing services than the market does.[1]

A voucher covers part of the rent, is means tested and is rationed. It is drawn independently each period, which is a simplification relative to the quantitative model, where a voucher persists as long as eligibility holds. Writing $\eta \in \{0,1\}$ for whether one is held,

$$\Pr(\eta' = 1 \mid y', d', v) \;=\; \pi_V \cdot \mathbf{1}\{y' \le y_V\} \cdot \mathbf{1}\{\text{policy covers } (d', v)\}, \tag{6}$$

and a household holding one pays the fraction κ_r of market rent, so the rent paid on option k is

$$r_k(\eta) \;=\; p_r\, h_k \big[\kappa_r \eta + (1-\eta)\big] \cdot \mathbf{1}\{k \neq 0\}, \tag{7}$$

which is zero when the household is homeless. Vouchers operate in both periods, and π_V, the share of eligible households actually covered, is the policy dial of Section 1.5.

The two policies differ in exactly one respect, the last indicator in equation (6). **Housing First** covers every eligible household regardless of (d', v). **Treatment First** covers only households that satisfy the sobriety condition, $v = 0$ and $d' = 0$, which is the rule in the quantitative model as well. Since policy coverage in the states that follow drug use is the only difference, the gap between them is the moral hazard and the whole of it.

The condition binds in period 1 as well, and the timing there is worth stating. Every household arrives clean, so the period 1 voucher is drawn before v is chosen and conditionality cannot screen anyone out on entry. The order within the period is therefore: the household chooses v, the policy rule is then applied to the voucher it drew, and it chooses housing knowing what rent it faces. So under Treatment First a household that uses would be subject to market rent immediately and again in period 2.

1.2.4 The household's problem

Let a be assets carried in. Period 2 is terminal, so the household consumes everything and chooses housing,

$$W_2(a,y,e,\eta,d) = \max_k \Big\{(1-\gamma-\theta)\log c + \gamma \log h_k + \aleph\Big\}, \qquad c = Ra + y^g(y,e) - r_k(\eta), \tag{8}$$

and write $W_2(a,y,d) \;=\; \mathbb{E}\big[W_2(a,y,e,\eta,d) \;\mid\; y,d\big]$ for its expectation over the period 2 employment and voucher draws. In period 1 the household draws (e,η) and then chooses housing, drug use and saving,

$$V_1(y,e,\eta) = \max_{k,\, v\ge 0,\, a'\ge 0} \Big\{(1-\gamma-\theta)\log c + \gamma \log h_k + \theta[\log(b+v) - \log b] + \aleph + \beta\, \mathbb{E}_{d'}\Big[\phi(d')\, W_2(a', y, d')\Big]\Big\}, \tag{9}$$

with $c = y^g(y,e) - r_k(\eta) - p_v v - a'$ and $d' = 1$ with the use dependent probability of (2). Survival sits inside the expectation over d' because it depends on which d' is realised.

A household uses if the value at its best positive v exceeds the value at $v = 0$. Households differ only in y and in the draw (e,η), which is the same lottery for everyone, so within a draw the decision defines a set of income intervals on which use occurs, and participation at a given income is the probability of drawing a state in which the household uses.

[1] Two rented sizes is the fewest that lets a household economise on housing rather than lose it. With one, homelessness comes out at 14.51 percent against a target of 0.40, at any h_1.

1.3 Calibration

1.3.1 External calibration

Table 1 lists the parameters set outside the model. Preferences and demographics are conventional: the discount factor over a thirty year period is $\beta = 0.860$[2] and the gross interest rate is $R = 1/\beta$, the dispersion of log permanent income is $\sigma_y = 0.75$ from Heathcote, Perri and Violante (2010), and survival over a period is $\phi_0 = 0.91$, from the Social Security Administration (SSA) period life table for twenty five to fifty five. For the SUD, the proportional fall in survival is $\kappa = 0.488$, derived by compounding an annual mortality ratio of 8, the midpoint of the range in Hjorthøj et al. (2023), so a clean household's chance of dying before period 2 is 9 percent and a household that acquires a SUD faces 48 percent, and the two unemployment probabilities are measured directly. $u_0 = 0.049$ is the headline civilian unemployment rate from the Bureau of Labor Statistics (BLS) Current Population Survey, series UNRATE as distributed by Federal Reserve Economic Data (FRED), averaged over 2014 to 2024. And $u_1 = 0.1625$ is from the National Survey on Drug Use and Health (NSDUH) 2024 for respondents aged 26 and over.

Income support replaces half of previous earnings on a base capped at mean income, in line with statutory state rules. The two rental sizes are set such that the standard unit costs 0.41 of mean income and the small unit is a quarter of it. The voucher parameters are institutional: covered tenants pay 30 percent of market rent, eligibility is capped at mean income, and $\pi_V = 0.27$ is the assisted eligible share in the 2023 Picture of Subsidized Households, published by the Department of Housing and Urban Development (HUD). The weight on drugs is set at $\theta = 0.20$.[3]

1.3.2 Internal calibration

Five parameters are set inside the model, in Table 2, each bracketed on the moment it moves most rather than fitted jointly. The weight on housing γ is matched to the 26.8 percent of total expenditure that shelter plus utilities take in the 2024 Consumer Expenditure Survey (CEX), housing over expenditure net of drugs. The flow value of being alive $\aleph$ is matched to a value of a statistical life of 80 times average annual income, about \$9 million against a mean household income of \$113,000, which is the range surveyed by Viscusi and Aldy (2003) once their estimates are put in current dollars. The pull of the first unit b is matched to a drug participation rate of 5.13 percent from NSDUH 2024 for respondents aged 26 and over, housing services on the street h_1 to a homelessness rate of 0.399 percent from the 2023 HUD point in time (PIT) count, converted to households, and the SUD hazard λ to the 33.2 percent of users who acquire a SUD.

1.4 Who uses drugs

This section asks which feature of the model puts drug use at the bottom of the income distribution, and it starts from the force that any answer has to overcome. Drugs are a luxury good bought at a price, so a household with twice the income gives up less utility per dollar spent on them. With no consequences at all for acquiring a SUD that is the only force operating, and drug use concentrates at the top: the average user then earns 2.89 times the population average.

Two features of the environment are candidates to reverse it. The *mortality channel* is that the SUD increases the chance of dying, through κ in equation (3). The *unemployment channel* is that it costs you a job with probability $u_1 - u_0$, and if income support cannot cover the rent you downsize, live on less, or lose your home. Figure 1 scales each from nothing to 125 percent of its calibrated value, recalibrating b at every point so participation stays at 5.13 percent. The amount of drug use is therefore identical all along both curves and the only thing moving is which households use.

[2]The annual discount rate here is higher than the one in the quantitative model, and this is deliberate. A period here is thirty years, so every consequence of using drugs arrives thirty years after the decision, whereas in the quantitative model consequences arrive annually and compound. A low annual discount factor would weigh that distant death much less and the mortality channel would be losing to impatience rather than to anything the model is about, so a higher factor corrects for the coarseness of the horizon rather than making a claim about preferences.

[3]A θ large enough to match the eleven percent of income that users spend on drugs makes the drug too valuable to be deterred by a death that arrives thirty years later, and users stop being poor. Setting $\theta = 0.20$ gives a spending share of 5.72 percent and keeps the income gradient, which is the trade this model can make.

Table 1: Externally set parameters

Parameter	Description	Source or rationale	Value
Preferences and demographics			
β	Discount factor over a thirty year period	0.99 annually, compounded	0.860
R	Gross interest rate over a thirty year period	Set to $1/\beta$	1.162
σ_y	Dispersion of log permanent income	Heathcote et al. (2010)	0.75
ϕ_0	Survival over a period, no SUD	SSA period life table, ages 25–55	0.91
θ	Weight on drug use	Set; see the text	0.20
The SUD			
κ	Proportional fall in survival with a SUD	Hjorthøj et al. (2023), age 26+, over 30 years	0.43
u_0	Unemployment probability, no SUD	BLS CPS, series UNRATE via FRED, 2014 to 2024	0.0485
u_1	Unemployment probability, with a SUD	NSDUH 2024, age 26+	0.1625
Income support			
ψ	Replacement rate while unemployed	DOL, statutory UI rates	0.50
$\bar{y}$	Cap on the earnings base, multiple of average income	DOL, state benefit maxima	1.00
Housing, drugs and the voucher			
p_r	Rent per unit of housing services	Normalisation	1.00
p_v	Price of a unit of drugs	Normalisation	1.00
h_3	Standard rental unit	Set	0.41
h_2	Small rental unit	Set	$0.25 \times h_3$
κ_r	Tenant's share of the rent when covered	HUD voucher rules	0.30
y_V	Voucher means test, multiple of average income	HUD income limits	1.00
π_V	Voucher generosity in the baseline	HUD 2023, assisted eligible share	0.27

Table 2: Internally calibrated parameters

Parameter	Description	Target moment	Data	Model	Value
γ	Weight on housing services	Housing over non drug spending	26.80%	27.03%	0.301
$\aleph$	Flow value of being alive	Value of a statistical life	80.0	80.0	2.899
b	Pull of the first unit of drugs	Drug participation rate	5.13%	5.13%	0.00226
h_1	Housing services when homeless	Homelessness rate	0.40%	0.43%	0.00630
λ	Addiction hazard per unit of use	Share of users who acquire a SUD	33.2%	37.1%	173.2

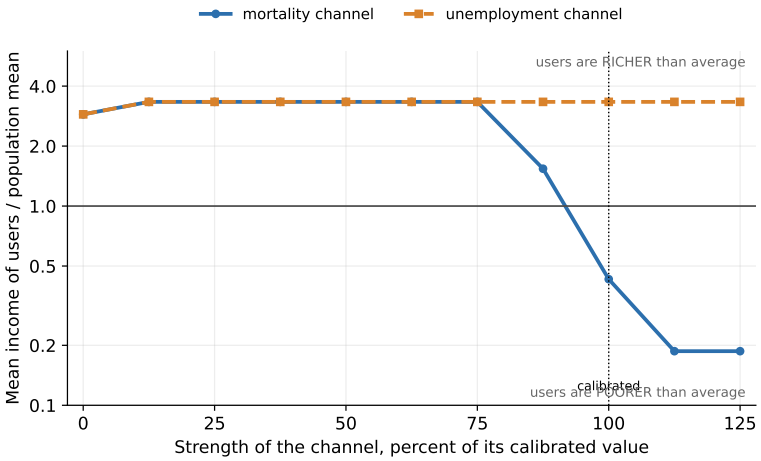


Figure 1: What it takes to reverse the income gradient in drug use

Note: the vertical axis is average income of users over the population average, on a log scale, and b is recalibrated at every point so that participation is 5.13 percent throughout. The black dotted line marks the calibrated strength; everything to the right of it is extrapolation.

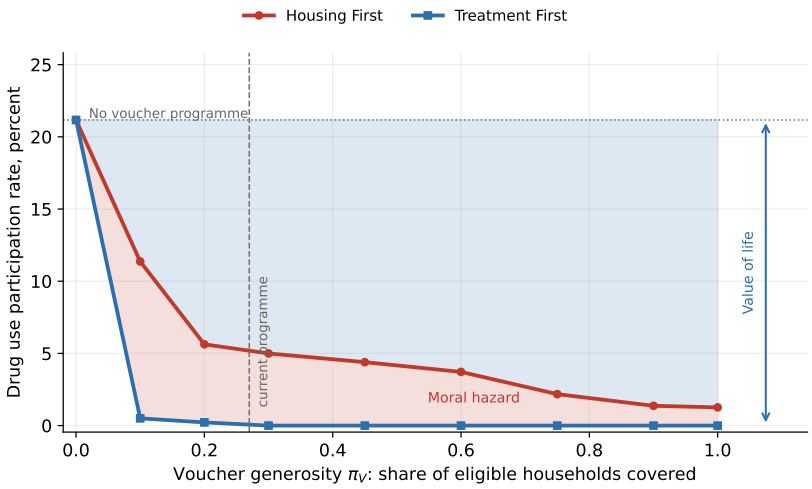


Figure 2: Drug use under the two policies as the voucher becomes more generous

Note: the red band is the moral hazard force and the blue band above it is what Housing First delivers net; together they are the value of life force, marked at the right. The dashed line is the generosity of the current programme, $\pi_V = 0.27$.

The result is sharper than the statement that mortality matters more. That channel takes the income ratio down to 0.43 at full strength, and it does so as a threshold rather than gradually: at half of its calibrated value the ratio is still high, at 3.33, and the reversal happens over the last tenth or so of the channel. Past the calibrated point it keeps biting, reaching 0.19 at 125 percent, so the reversal is not a knife edge that the calibration happens to sit on. The unemployment channel does not reverse it at all, leaving users at 3.33 times average income even fully switched on, and at 125 percent it is 3.33, still above one and no closer to turning. Strengthening the resource penalty by a quarter does not begin to do what mortality does. A resource penalty does not produce the concentration of drug use at the bottom.

A mortality penalty is different in kind. A resource penalty scales with the very thing it is meant to offset, since taking a fixed share of a large income still leaves a large income, so it cannot outrun the fact that drugs are cheaper in utility terms for the rich. What mortality destroys is not resources but a continuation value, and the continuation value rises with permanent income. The household with more to lose loses more, and loses proportionally more than it gains from drugs being relatively cheap. That is the work of $\aleph$ in equation (1): it is the part of the continuation value that does not shrink as consumption falls.

1.5 How does housing assistance shape the drug use decision?

The two forces of question 2 can be separated exactly, because they differ in what happens to a household that uses. Under Treatment First the voucher is withdrawn from any household that violates the sobriety condition, so only the value of life force operates. Under Housing First it is not, so both channels operate. Comparing the two policies at the same generosity therefore isolates the moral hazard force, and comparing Treatment First with no programme at all isolates the value of life force.

Figure 2 traces both as generosity rises. With no programme at all, 21.17 percent of households use. Universal Treatment First takes that to 0.00 percent, a value of life force of -21.17 percentage points; universal Housing First leaves it at 1.26 percent, so the moral hazard force is +1.26 points and the net effect of Housing First is -19.91. Both forces are first order and they work against each other, which is the central feature of the problem. Anyone asserting that housing the homeless must increase drug use, or must reduce it, is making a quantitative claim that needs to be tested.

Figure 3 plots the homelessness and SUD outcomes for each programme as its generosity changes, alongside drug use. In panel (a) the two policies end in the same place on homelessness, both driving it from 0.52 percent with no programme to zero, and along the way Treatment First is if anything slightly below Housing First, because the households it deters from using never acquire the SUD that puts their housing

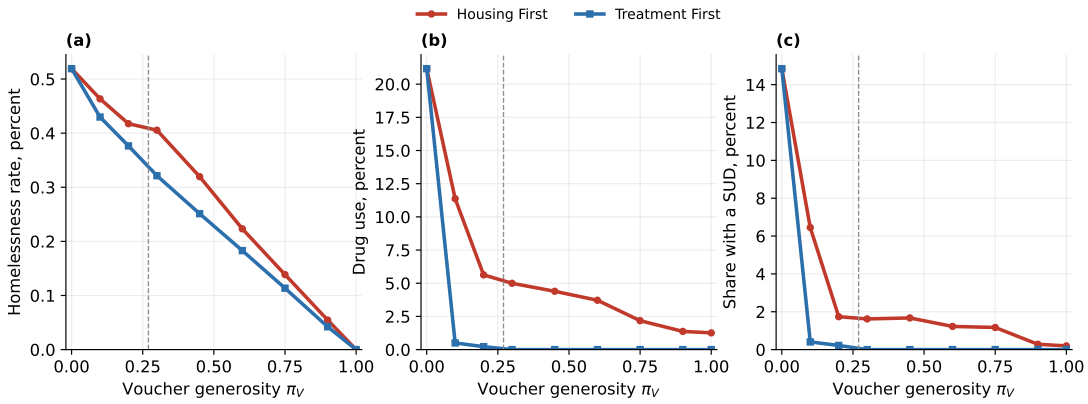


Figure 3: Homelessness, drug use and the SUD under the two policies

at risk. Panel (b) repeats Figure 2 here for convenience. The larger difference is in panel (c), where at universal cover Housing First leaves 0.196 percent of the population carrying the SUD and Treatment First leaves 0.000 percent. What the moral hazard buys is extra SUD cases among households that are housed either way.

1.6 Welfare

1.6.1 Without externalities, Housing First always wins

Sections 1.4 and 1.5 establish what each policy does to behaviour. This section asks what each does to welfare, comparing the two at the same generosity π_V, whose outcomes are in Panel A of Table 3. The policies differ only in the last indicator in equation (6), so Housing First transfers resources in a superset of the states in which Treatment First does. The budget set is weakly larger in every state and strictly larger in one, so a household that optimises cannot be worse off under Housing First:

$$V_1(y;\ \text{HF},\ \pi_V)\ \geq\ V_1(y;\ \text{TF},\ \pi_V) \qquad \text{for every } y, \tag{10}$$

with equality exactly for households that never use. Put the other way round, a household under Housing First can reproduce its Treatment First outcome exactly by not using: conditionality removes an option from a forward looking optimiser and gives nothing back. This is a dominance argument, not a calibrated result, and no choice of parameters overturns it.[4]

It follows that the moral hazard is not a distortion. Households here face the true SUD hazard and make no mistakes, so the extra use Housing First induces is not a wedge to be closed but a household optimally taking a risk it is now better able to bear. The +1.26 points of Section 1.5 are a statement about drug use and not about welfare. Panel B of Table 3 puts numbers on it: 2.7 percent of households are strictly worse off under Treatment First, none is better off, and those affected lose 0.45 percent of consumption. Those are exactly the households that use: a household that never uses faces the same voucher rule under both policies and is indifferent between them. The figure is not the 5.13 percent participation rate of Section 1.3 because the programmes compared here are universal, rather than the rationed benchmark that rate is measured at.

A fixed budget does not rescue the welfare case for conditionality either. Treatment First is exactly as expensive as Housing First at universal cover, 0.3185 against 0.3185, so matching on cost rather than on generosity changes nothing. Two things offset. Survival into period 2 falls from 0.91 to 0.52 with a SUD, so the extra households Housing First puts into the SUD state die sooner and fewer are alive in period 2 to collect a voucher; against that, Treatment First withdraws the voucher from anyone who used, so it pays for fewer tenants while they live. The expected outlay remains the same. So a case for conditionality has to come from an externality.

[4]The same comparison in the quantitative model gives the same answer.

Table 3: Welfare at universal cover, with no externality

	Housing First	Treatment First
Panel A. Outcomes at universal cover		
Drug participation rate	1.26%	0.00%
Share with a SUD	0.196%	0.000%
Homelessness rate	0.000%	0.000%
Voucher outlay per household	0.3185	0.3185
Panel B. Treatment First against Housing First, no externality		
Households strictly worse off		2.7%
Households strictly better off		0.0%
Consumption equivalent loss, those affected		0.45%
Consumption equivalent loss, whole population		0.012%

Note: the consumption equivalent is the proportional change in consumption, in every period and state, worth as much as the difference in lifetime value. Voucher outlay is expected spending on a household over both periods, the second discounted and weighted by survival, per unit of average income.

Table 4: The externality on children

Panel A. Transmission: what each income penalty implies			
δ_d	Child's SUD rate	Ratio	Loss to the child
0.00	2.24%	1.00	0.0%
0.20	4.04%	1.80	38.7%
0.30 (calibrated)	5.45%	2.43	69.2%
0.40	7.43%	3.31	113.3%
Panel B. Does the externality reverse the ranking?			
Private welfare loss from conditionality			0.012%
Disorder cases prevented			0.196 pp
Damage to the child of one such case			54%
Externality gain from conditionality			0.106%
Gain over loss			8.9

Note: Panel A is at the baseline programme, $\pi_V = 0.27$, since the transmission ratio in the data comes from an economy with a rationed programme; Panel B is at universal cover, one child per household, no discounting across generations. Losses are consumption equivalents.

1.6.2 An externality on children reverses it

I model an externality that runs through income rather than through preferences. I have considered two alternatives, but both seem to be at odds with reality. Letting the child of a user inherit a stronger taste, a lower b, matches the transmission fact but has the wrong welfare sign: b enters only through $\theta \log(1 + v/b)$, so a lower one raises the child's objective at every $v > 0$ and its lifetime value with it. So the specification says that a child who inherits a stronger taste for drugs is *made better off* by inheriting it, and that is the opposite of what these children report. Growing up with household substance abuse is one of the adverse childhood experiences in Felitti et al. (1998), and it predicts higher rates of depression, worse self reported health and lower life satisfaction in adulthood; the children of parents with a SUD also have lower educational attainment, which is the premise of the transmission literature δ_d is calibrated against (McGovern et al., 2023). A model in which the transmitted trait raises the child's own welfare cannot be describing any of that, whatever it does to the transmission rate. Letting it inherit a greater susceptibility, a higher λ, fails another way, since a child facing a riskier drug optimally uses *less* ex ante, which is at odds with the data: children of parents with a SUD are *more* likely to use, not less.

Income works on both counts, and it works through the channels explored in Section 1.4: lower permanent income is unambiguously bad to receive, and this model already puts users at the bottom, so a child born poorer is mechanically more likely to use, which is aligned with the data. It is also supported by the evidence, and it stands in for more than income alone: a parent who diverts spending and time to drugs invests less in the child's education and development, and the income penalty is the reduced form of all of that (Chetty, Hendren and Katz, 2016; McGovern et al., 2023). The income gradient does the whole job, so the model

transmits a SUD across generations *without transmitting any preference for drugs.*

A child's baseline income is drawn from the same lognormal as everyone else, and the entire link is the penalty

$$y_{\text{child}} \;=\; y_{\text{base}}\,(1-\delta_d)^{\mathbf{1}\{\text{parent acquired the SUD}\}}\,(1-\delta_h)^{\mathbf{1}\{\text{parent was homeless}\}}, \tag{11}$$

which the parent does not internalise. That is what makes it an externality rather than something already priced into V_1.[5]

The two penalties are set to different things. Panel A of Table 4 shows that $\delta_d = 0.30$ makes the child of a parent who acquired the SUD 2.4 times as likely to acquire one, in the middle of the epidemiological range of two to four in McGovern et al. (2023) and the alcohol literature. The homelessness penalty δ_h is set to 31 percent, directly from Chetty et al. (2016), who find that children who moved to a low poverty neighbourhood before age 13 went on to earn 31 percent more as adults. The two penalties also do different work: both policies drive homelessness to zero at universal cover, so δ_h prices the programme against no programme while δ_d decides Treatment First against Housing First.

Panel B puts the two sides together and they are not close. The private cost of conditionality, in the first row, is 0.012 percent of consumption, small because only 2.7 percent of households are affected. Against that, conditionality prevents 0.196 percentage points of SUD cases in the second row, and each of those cases costs one child 54.3 percent of consumption, shown in the third row. The gain in the fourth row is the product of those two, since one prevented case spares one child: 0.196 percentage points times 54.3 percent gives 0.106 percent of consumption, which is 8.9 times the private loss of the first row. That ratio exceeds one for every δ_d in Panel A except $\delta_d = 0$, where by construction there is no externality to price. Housing First is better for every household taken one at a time, and worse once one counts the children of the households it fails to deter.

All of this is a crude illustration: the model is two periods long and nobody who faces the sobriety condition fails to meet it, so the impacts of policies can be unreasonably large. It shows that an externality can reverse a welfare ranking that is otherwise airtight, which is all it is for. Which way the comparison actually goes is a question about magnitudes, and that is what the quantitative model of the next section is for.

The quantitative model also carries more externalities than are described here, and they do not all pull the same way, which is another substantive reason the welfare ranking is a question of magnitudes rather than of signs.

- **Programmes are paid for out of distortionary taxes, and they shrink the base they are paid from.** A programme that houses more households costs more, raising taxes that distort the hours and the saving of every household, including those it never reaches. It also lowers the labour supply of the households it does reach, because eligibility is means tested: the voucher is clawed back at a rate of κ_r, so a recipient who works more loses some of the transfer. This means that other households have to pay more taxes, which further distorts their labour supply. Neither effect exists here, where hours are fixed and financing is not modelled.
- **Homelessness is a disamenity in itself.** Households dislike homelessness around them and not only their own, so the aggregate rate is an argument of everyone's utility. It runs in favour of whichever policy houses more, which is the opposite direction from the tax.

2 The quantitative model

This section describes the model the paper's numbers come from. Where the stylized model already carries something unchanged it is named rather than motivated a second time: the Stone-Geary form of the drug term, the logic that separates the two policies, and the transmission externality all appear below. Notation follows Section 1 wherever the two models share an object: γ and θ are the housing and drug weights in both, b is the same Stone-Geary endowment, $\aleph$ the same flow value of being alive, λ the same SUD hazard,

[5] In the quantitative model the altruism of parents towards their children is calibrated rather than set to zero, so that the channel where a parent does take some account of what it does to its child is present and only the remainder is external.

ψ and $\bar{y}$ the same replacement rate and earnings base cap, r the same risk free rate, and β is the discount factor and nothing else.

2.1 Demographics, the state and the choice set

Households enter as adults, live at most J annual periods and retire at J_R, and j indexes age within that span. Survival from j to $j+1$ is $\phi(j, h', d, m)$ in (24), and it depends on age, on the housing *chosen* this period, on the SUD and on serious mental illness (SMI). Newborns replace the dead exactly, so the mass of households is constant. The individual state is

$$s = (a,\, y,\, e,\, d,\, m,\, \eta) \in \mathcal{A} \times \mathcal{Y} \times \{1,2,3\} \times \{0,1\}^3, \tag{12}$$

where a denotes assets, y permanent productivity, e employment status, d the SUD, m SMI and η receipt of a voucher.

Each period the household chooses

$$\big(n,\, h',\, v,\, \tau,\, a'\big) \in [0,1] \times \{h_1, h_2, h_3, h_4\} \times \mathbb{R}_+ \times \{0,1\} \times \mathcal{A}, \tag{13}$$

where n denotes hours, h' the housing unit chosen, of which h_1 is homelessness, v drug use, τ SUD treatment take up, available only when $d = 1$ and requiring $v = 0$ in the period it is taken, and a' next period assets.

2.2 Preferences

Period utility is

$$u\big(c, h', v, n, \tau; d, m\big) \;=\; \frac{\big[\, c^{1-\gamma-\theta}\,(h')^{\gamma}\,(v+b)^{\theta}\,\big]^{1-\sigma}}{1-\sigma} \;+\; \omega_\ell \frac{(1-n)^{1-\sigma_\ell}}{1-\sigma_\ell} \;+\; \aleph \;-\; \tau\bar{\tau}^{u} \;-\; \kappa_x H, \tag{14}$$

Four of these are the stylized model's and carry the same meaning here as in Section 1.2: γ the housing weight, θ the drug weight, so that $1-\gamma-\theta$ is the weight on consumption and the three sum to one; b the Stone-Geary endowment; and $\aleph$ the flow value of being alive. Equation (1) is the logarithmic case of this bundle with hours fixed.

The rest are new, and there are three of them. Leisure makes hours a margin, at weight ω_ℓ and curvature σ_ℓ over the $1-n$ hours not worked. SUD treatment costs utility, $\bar{\tau}^u$ in each period the household takes it up, $\tau = 1$. And homelessness carries a disamenity, κ_x per unit of the aggregate homelessness rate H, borne by every household and priced on the headcount alone, which is what makes homelessness an externality here rather than only a private misfortune; Section 3.4 takes it up. Outside the bundle, σ is the coefficient of relative risk aversion.

2.3 Income, taxes and transfers

Gross income depends on age, employment, productivity and health,

$$y^{g}(j,y,e,d,m,n) = \begin{cases} 0.40\, y & j \geq J_R \quad \text{(social security)} \\ n\, y\, \varepsilon(j)\, \rho_d^{\mathbf{1}\{d=1\}} \rho_m^{\mathbf{1}\{m=1\}} & j < J_R,\ e = 1 \quad \text{(employed)} \\ \psi\, \min\{y\, \varepsilon(j),\ \bar{y}\} & j < J_R,\ e = 2 \quad \text{(first year of a spell)} \\ \zeta_{DI}\, y\, \varepsilon(j) & j < J_R,\ e = 3,\ m = 1 \quad \text{(non employed with SMI: disability)} \\ 0 & \text{otherwise,} \end{cases} \tag{15}$$

taking the cases in the order they appear. Retirement income replaces 40 percent of the productivity state with no health penalty, since social security is not means tested on health. For the employed, $\varepsilon(j)$ is the deterministic age profile of efficiency and ρ_d and ρ_m are productivity penalties applied *conditional on working*. The three employment states are work, $e = 1$, the first period of a spell out of work, $e = 2$, and

non employment thereafter, $e = 3$, so the last two cases are both unemployment and the disability case is the income of a household that is out of work and carries SMI. Unemployment insurance (UI) is paid in the first period of a spell only, at the replacement rate ψ on an earnings base capped at average income $\bar{y}$, both as in equation (4). ζ_{DI} is the rate at which disability insurance (DI) replaces the income of a non employed household with SMI.

The two penalties, ρ_d and ρ_m, are in the model to generate a large share of homeless households being employed, consistent with the data. Employed homelessness cannot be produced by unemployment alone: something has to leave a household poor while it is still in a job. Both are calibrated internally: ρ_d on the share of the homeless who are employed, and ρ_m on the earnings of workers with SMI relative to average.

Labour income is taxed with the Heathcote et al. (2017) function and capital income at a flat rate τ_k,

$$y^n = \tau_\ell \, (y^g)^{\xi} \; + \; (1 - \tau_k) \, r \, a, \tag{16}$$

with y^n denoting after tax income, where ξ governs progressivity and τ_ℓ the level. Households can receive two transfers, both means tested on gross income as the programmes themselves are. A food stamp floor tops means tested income up to $\underline{c}\,\bar{y}$, with $\underline{c}$ the floor expressed as a share of average income, and the voucher covers rent above a tenant contribution of κ_r of gross earnings, the same κ_r as in Section 1.2.3,

$$F = \max\big\{0, \; \underline{c}\,\bar{y} - (y^g + ra)\big\}, \qquad Vch = \mathbf{1}\{\eta = 1\}\,\mathbf{1}\{h' \neq h_1\}\,\max\big\{0, \; p_r\,h' - \kappa_r\,y^g\big\}. \tag{17}$$

A homeless voucher holder does not lose the voucher: retention depends on income eligibility, not on housing, so the entitlement survives and simply pays nothing until the household rents again.

2.4 The budget constraint

$$c + p_v v + a' \;=\; y^n + a \;-\; p_r\,h'\,\mathbf{1}\{h' \neq h_1\} \;+\; Vch + F, \qquad c > 0, \; v \geq 0, \; a' \geq 0. \tag{18}$$

The drug price is p_v and market rent is p_r, both as in Section 1.2.3, and a is assets carried in. Note what is absent: treatment, which costs utility, and bequests, which go to the government. How c, v and n are chosen out of it is with the rest of the optimisation, in Section 2.11.

2.5 Housing, and the homelessness loop

Housing is a four point grid. h_1 is literal homelessness, sleeping rough or in an emergency shelter, which is the HUD point-in-time definition and excludes doubling up; it carries a small service flow and no rent. The three rental sizes are geometric,

$$h_2 = \underline{h}, \qquad h_3 = \underline{h}\,\varsigma, \qquad h_4 = \underline{h}\,\varsigma^2, \tag{19}$$

so two parameters describe the grid: $\underline{h}$ the smallest rental size and $\varsigma > 1$ the geometric spread between consecutive sizes.

Homelessness then enters three separate equations: survival (24), the employment transitions (26) and the SUD recovery hazard (21), through $\kappa_{\phi H}(j)$ and κ_{ueH} in Section 2.8 and μ_H in Section 2.6. It raises mortality, worsens both employment margins and lowers the chance of recovering from the SUD. Mental illness is the one thing it does not reach. The SMI chain runs on its own, and that is a finding rather than an omission: the trial that housed people with serious mental illness looked for an effect on their symptoms and did not find one, while finding large effects on their housing. Section 2.7 gives it.

2.6 Drug use, the SUD, recovery and treatment

A household either abstains or uses, with v chosen continuously. A clean household that uses acquires the SUD with the hazard of equation (2),

$$\Pr(d' = 1 \mid d = 0, v) = \pi_d(v) = 1 - e^{-\lambda v}, \tag{20}$$

In the baseline a clean household may initiate only up to an age J_E, the last age at which use can begin.[6] A household with the SUD recovers with probability

$$\pi_r(h', m, \tau) = \min\left\{1,\ \bar{\pi}_r\, \mu_H^{\mathbf{1}\{h'=h_1\}}\, \mu_m^{\mathbf{1}\{m=1\}}\, g^{\tau}\right\}, \tag{21}$$

with $\bar{\pi}_r$ the baseline annual chance of recovering while housed, clean of SMI and untreated, and three multipliers on it: $\mu_H < 1$, so recovery is harder on the street, $\mu_m < 1$, harder with SMI, and $g > 1$ the treatment effect.

μ_H is how housing acts on drug use here, and it is worth being exact about what that claim is and is not. Housing does not change what a household wants: flow utility is the same function of (c, h, v) housed or not. What housing changes is the chance of getting out of the disorder once it is carried, recovery on the street being μ_H times recovery while housed. Milby et al. (2005) randomised cocaine dependent homeless adults into arms differing in housing and measured weeks of abstinence, so the ratio between arms identifies μ_H in our calibration.

Treatment τ is a *choice*, not a mechanical hazard, taken at the utility cost $\bar{\tau}^u$ in (14). It also costs the drug: a household in treatment must abstain in that period, $v = 0$ whenever $\tau = 1$. And g is the parameter through which its entire effect runs, so treatment has to be something a household decides to undertake and can decline.

2.7 Serious mental illness

m is an exogenous two state Markov chain representing SMI, whose prevalence p_m is about 5.5 percent of adults, defined by substantial functional impairment, rather than any mental illness, which is roughly a quarter of adults and includes mild anxiety: a state that halves productivity cannot sensibly cover a quarter of the population.

The onset rate is a function of age. Writing $p(j)$ for prevalence at age j and λ_m for the annual chance of leaving the state,

$$\text{onset}_m(j) = \frac{p(j{+}1) - p(j)(1 - \lambda_m)}{1 - p(j)}, \tag{22}$$

which is the accounting identity that reproduces the observed age profile exactly, backed out of the age profile of prevalence in NSDUH 2023 and the recovery hazard. The transitions are then

$$\Pr(m' = 1 \mid m = 0, d) = \text{onset}_m(j) \cdot \iota^{\mathbf{1}\{d=1\}}, \qquad \Pr(m' = 0 \mid m = 1) = \lambda_m, \tag{23}$$

where $\iota > 1$ is the multiplier on onset for a household carrying the SUD, which makes a SUD raise the hazard of SMI. That last channel runs in the direction the evidence supports: substance induced psychosis converts to schizophrenia or bipolar disorder at a pooled rate near a quarter (Murrie et al., 2020), and in the Danish registers any SUD predicts conversion to schizophrenia with a hazard ratio of 1.34, opioid use disorder 2.74 (Nielsen et al., 2017).

SMI then affects five objects: productivity conditional on working through ρ_m in (15), both employment margins through κ_{uem} in (26), mortality through $\kappa_{\phi m}$ in (24), the SUD recovery hazard through μ_m in (21), and disability income, for which eligibility is being out of work with SMI and the replacement rate is ζ_{DI}. It does not change the utility function.

The mental health chain does *not* depend on housing, and that is a finding in the data rather than an omission. In the At Home / Chez Soi randomised trial, 1,198 homeless adults with SMI across four cities, assignment to housing with intensive case management produced no detectable effect on symptom severity at any wave, while producing large effects on housing stability, its primary outcome. Making the channel housing dependent would mean adding a mechanism the best available trial looked for and did not find.

[6] Without it there is little consequence at old ages to using: too little life remains for the disorder to damage, so the model produces too much of it.

2.8 What the SUD, SMI, and homelessness cost

Death risk. Survival is

$$\phi(j,h',d,m) = 1 - \min\Big\{1,\ (1-\bar{\phi}(j))\,\kappa_{\phi H}(j)^{\mathbf{1}\{h'=h_1\}}\,\kappa_{\phi d}(j)^{\mathbf{1}\{d=1\}}\,\kappa_{\phi m}^{\mathbf{1}\{m=1\}}\Big\}, \tag{24}$$

with $\bar{\phi}(j)$ the baseline survival age profile and $\kappa_{\phi H}(j)$ and $\kappa_{\phi d}(j)$ two age varying multipliers, and $\kappa_{\phi m}$ one that is not, raising the death hazard for homelessness, the SUD and SMI respectively. Because $\aleph > 0$ and guarantees that the value of life is positive, a higher death hazard is a real cost, and it is larger for households with more life remaining and more to consume, which is precisely the sorting Section 1.4 isolates.

$\bar{\phi}(j)$ is the SSA period life table for ages 26 to 85, with the sexes combined by survivor weight, $q = [l^M q^M + l^F q^F]/[l^M + l^F]$.

The homelessness multiplier cannot be read off the data directly, and the reason is worth spelling out. Baggett et al. report a *standardised mortality ratio* (SMR) for homeless adults: deaths among the homeless at a given age divided by deaths in the general population at that age, which they publish by age band and which we write $RR(j)$. It cannot be used as $\kappa_{\phi H}(j)$ because the people it is measured on are not only homeless. A share $s = 0.217$ of them also carry a SUD (Gutwinski et al., 2021), which raises mortality by itself, so the published ratio is a mixture of two effects and attributing all of it to housing would overstate what housing does.

Three things are observed and two are solved for. Observed: $RR(j)$, Baggett's ratio; $\sigma_d(j)$, the shape of the SUD's mortality effect over age; and s, the SUD share of the homeless. Solved for: $\kappa_{\phi d}(j)$ and $\kappa_{\phi H}(j)$, one age profile from each equation. Picking the single number $\bar{\kappa}_{\phi d}$ settles both: the first equation turns it into $\kappa_{\phi d}(j)$ at every age, and the second then reads $\kappa_{\phi H}(j)$ off Baggett's ratio at every age.

Undoing the mixture needs the SUD's own multiplier at each age. That is what $\sigma_d(j)$ supplies: the age profile of drug poisoning deaths from CDC WONDER 2023, normalised to one at the age where it peaks, so $\sigma_d(j)$ is the share of the SUD's peak mortality effect that is present at age j. Writing $\bar{\kappa}_{\phi d}$ for the level of the SUD multiplier,

$$\kappa_{\phi d}(j) = 1 + (\bar{\kappa}_{\phi d} - 1)\,\sigma_d(j), \qquad \kappa_{\phi H}(j) = \max\Big\{1,\ \frac{RR(j)}{(1-s) + s\,\kappa_{\phi d}(j)}\Big\}. \tag{25}$$

The denominator is the average mortality multiplier in Baggett's population: a share $1-s$ of it carries homelessness alone and a share s carries the SUD as well. Dividing the observed ratio by it leaves the homelessness effect on its own.

$\bar{\kappa}_{\phi d}$ is assigned from the literature on all cause standardised mortality ratios for opioid use disorder, which report values between 6 and 10 (Degenhardt et al., 2011; Hser et al., 2017; Larney et al., 2020), and the calibration section reports the value taken. The drug price participation elasticity is not targeted; it is reported as a validation moment and compared with the range estimated in the literature, -0.3 to -1.1 (Saffer and Chaloupka, 1999; Grossman and Chaloupka, 1998; Olmstead et al., 2015).

$\kappa_{\phi m}$ carries no age argument because the data give it basically none. Olfson et al. (2015) give all cause standardised mortality ratios for schizophrenia of 3.6, 3.4 and 4.0 across ages 20 to 34, 35 to 54 and 55 to 64. Flat, if anything rising.

Unemployment risk. Job finding and separation are

$$\pi^{ue} = \min\Big\{1,\ \bar{\pi}^{ue}\,\kappa_{ueH}^{\mathbf{1}\{h'=h_1\}}\kappa_{ued}^{\mathbf{1}\{d=1\}}\kappa_{uem}^{\mathbf{1}\{m=1\}}\Big\}, \qquad \pi^{eu} = \min\Big\{1,\ \bar{\pi}^{eu}\,\kappa_{ueH}^{-\mathbf{1}\{h'=h_1\}}\kappa_{ued}^{-\mathbf{1}\{d=1\}}\kappa_{uem}^{-\mathbf{1}\{m=1\}}\Big\}, \tag{26}$$

with $\bar{\pi}^{ue}$ and $\bar{\pi}^{eu}$ the baseline job finding and separation rates and κ_{ueH}, κ_{ued} and κ_{uem} the multipliers for homelessness, the SUD and SMI, so each multiplier below one makes work simultaneously harder to find and easier to lose.

The child's income risk. A parent with a SUD, or one who ends up on the street, lowers the productivity (y_{child}) its child is born with, through δ_d and δ_h respectively, as shown in (39) in Section 2.14. That is a risk

to the household's own objective only to the extent that ω, the weight the parent places on its child's lifetime utility in (30), is positive. Since ω is small (see Section 3 for details), the household weighs a fraction ω of this risk and $1-\omega$ of it falls outside the problem entirely. It is therefore the one risk here that is principally an externality rather than a pure personal cost, which is what makes it a candidate for policy in a way that mortality and unemployment risk, both fully internalised, are not.

2.9 The voucher, and the two policies

A household without a voucher draws one with probability π_V, the same lottery probability as in Section 1.2.3, if it is income eligible, and a household with one keeps it while eligible, so unlike Section 1.2.3 the voucher is persistent rather than redrawn each period. Eligibility is

$$y^g + ra \;<\; \kappa_V \, y^{\text{med}}. \tag{27}$$

with κ_V the eligibility cutoff expressed as a share of median gross income y^{med}. Under Housing First that is the whole rule. Under Treatment First the voucher is additionally lost with probability p_{det} by a household that uses drugs or has a SUD, and with probability p_{fp} by one that did not,

$$\Pr(\eta'=1) = \Pr(\eta'=1)^{HF} \times \begin{cases} 1-p_{\text{fp}} & \text{if } (v=0 \text{ and } d=0) \text{ or } \tau=1, \\ 1-p_{\text{det}} & \text{otherwise.} \end{cases} \tag{28}$$

There are four cases and the rule covers them. A clean household that does not use meets the condition and faces only the false positive rate p_{fp}. A clean household that uses fails it and faces p_{det}. A household carrying the disorder that does not take treatment fails it and faces p_{det} as well. And a household carrying the disorder that does take treatment is abstaining, $v=0$ by the restriction in Section 2.6, so it meets the condition and faces p_{fp}. There is no fifth case: treatment is available only when $d=1$, and it forbids use, so no household is ever both in treatment and using.

This is the one place where the quantitative model changes the *logic* of Section 1.5 rather than its setting. There, conditionality is perfectly enforced and perfectly met, so Treatment First deters everyone and nobody is thrown off the rolls. Here detection is imperfect and so are households: some fail the condition because they cannot get clean, and some fail it because the test is wrong. Those two groups are the empirical case for Housing First, neither exists in the stylized model, and that is why its welfare numbers were illustration.

2.10 The shocks, collected

Conditional on the state and the choices, next period's state is drawn from five independent transitions. Collecting them in one place:

Productivity y. Fixed over the life cycle, so $y'=y$. I do this for two reasons. First, it is what keeps the child's penalty (39) a shift along a single scalar rather than a shift across the nodes of an age varying discretisation, which significantly reduces the computation time to manageable levels. Second, it still captures the key mechanisms, because unemployment and the penalty ρ_d are the income risks that link the SUD to housing, and those are retained in full.

Employment e. A household that loses their job receives insurance with certainty for one year. With π^{ue} and π^{eu} from (26), it gives the following transitions:

$$e=1 \;\longrightarrow \begin{cases} e'=1 & \text{w.p. } 1-\pi^{eu} \\ e'=2 & \text{w.p. } \pi^{eu} \end{cases} \qquad e\in\{2,3\} \;\longrightarrow \begin{cases} e'=1 & \text{w.p. } \pi^{ue} \\ e'=3 & \text{w.p. } 1-\pi^{ue}, \end{cases} \tag{29}$$

State 2 is reachable only from employment and never from itself, so a household is in it for the first year out of work and in state 3, which pays nothing, from the second year on. One year is if anything generous: most states pay for twenty six weeks.

The SUD d. By (20) if $d = 0$ and (21) if $d = 1$, conditioning on the chosen v, h' and τ.

Mental illness m. By (23), conditioning on d.

The voucher η. By (27) and (28), conditioning on next period's eligibility and, under Treatment First, on the chosen v, the status d and the treatment choice τ.

Write $\Gamma\big(s' \mid s,\, n, h', v, \tau\big)$ for the product of these five, which is the object the expectation in (30) is taken against.

2.11 The household's dynamic problem

Putting it together, for $j = 1, \ldots, J$ and with $V(J{+}1, \cdot)$ given below,

$$
\begin{aligned}
V(j,a,y,e,d,m,\eta) = \max_{\{n,\,h',\,v,\,\tau,\,a'\}} \Bigg\{ & \frac{\big[c^{1-\gamma-\theta}(h')^{\gamma}(v+b)^{\theta}\big]^{1-\sigma}}{1-\sigma} + \omega_{\ell}\frac{(1-n)^{1-\sigma_{\ell}}}{1-\sigma_{\ell}} + \aleph - \tau\bar{\tau}^{u} - \kappa_x H \\
& + \ \beta\,\phi(j,h',d,m)\int_{\mathcal{S}} V(j{+}1)(s')\,\Gamma\big(\mathrm{d}s' \mid s,n,h',v,\tau\big) \\
& + \ \beta\,\big[1-\phi(j,h',d,m)\big]\,\omega\,\mathbb{E}\,V\big(1,\,0,\,y_{\text{child}},\,e',\,0,0,0\big) \Bigg\}
\end{aligned}
\tag{30}
$$

subject to the budget (18), the income definitions (15) to (17), the choice set of dwellings (13), $\tau = 0$ whenever $d = 0$, $n = 0$ whenever $j \geq J_R$ or $e \neq 1$, $v = 0$ whenever $\tau = 1$, $a' \geq 0$, $c \geq F$ so that consumption never falls below the food stamp floor, and $V(J{+}1, \cdot) = 0$ with $\phi(J) = 0$, so that at the last age the child is the whole continuation. The integral runs over next period's state $s' = (a', y, e', d', m', \eta')$; the second line of (30) is the household's own continuation, and the third is the child's expected lifetime utility, entered at age one with zero assets and with its own employment status e' drawn as any newborn's is.

A household that does not survive from j to $j + 1$ leaves a child, so the continuation is the survival weighted value of its own next period plus the death weighted value of the child's life, each discounted, with y_{child} as defined in (39). What that buys is a second route by which the household's own decisions, to use and to go unhoused, reach its child. $\phi(j, \cdot)$ itself depends on h' and, through d, on v: a household that uses raises the chance it is the child's term rather than its own that is realised, so the child enters the drug and housing decision through the weight on the two terms as well as through y_{child}.

At $\omega = 0$ death is worth zero and the damage a household does to its child is entirely external; at $\omega > 0$ the household internalises a share ω of it, and only $1 - \omega$ remains external. Since the child's productivity depends on whether the parent carried a SUD or was homeless, and both depend on v and h', the child term in (30) is what puts the child inside the parent's drug and housing decision.

How the choices separate, and where they do not. Taking hours, the dwelling and next period assets as given, what is left to divide between consumption and the drug is

$$
x \;=\; y^{n} + a \;-\; p_r\, h'\, \mathbf{1}\{h' \neq h_1\} \;+\; Vch + F \;-\; a', \tag{31}
$$

so the budget (18) reads $c + p_v v = x$ and the whole of the remaining problem is one number: how to split x between c and v, since either one determines the other.

Were the drug an ordinary good, one whose only consequence is the flow utility it delivers this period, that split would follow from flow utility alone. It does not, because v enters the transition as well as the utility: by (20) another unit of the drug also buys a higher chance of waking up next period with the disorder. So there are three terms in the first order condition for v, not two,

$$
\underbrace{\frac{\theta}{b+v}\,\Lambda}_{\text{value of another unit}} \;=\; \underbrace{\frac{p_v\,(1-\gamma-\theta)}{c}\,\Lambda}_{\text{the consumption it displaces}} \;+\; \underbrace{\lambda e^{-\lambda v}\Big[\,\mathbb{E}V(\cdot \mid d'=0) - \mathbb{E}V(\cdot \mid d'=1)\,\Big]}_{\text{the risk it adds}}, \tag{32}
$$

where $\Lambda = \left[c^{1-\gamma-\theta}(h')^{\gamma}(v+b)^{\theta}\right]^{1-\sigma}$ is the Cobb-Douglas bundle of (14). It multiplies the first two terms and not the third, because the third is a difference of continuation values and is therefore already in utils. Were it absent, Λ would divide out of the condition entirely and the split would reduce to the usual Cobb-Douglas one; it is the third term that stops it cancelling, and that is why the marginal utility level matters here and not in an ordinary two good problem. That third term is the marginal hazard $\pi_d'(v) = \lambda e^{-\lambda v}$ times what the disorder costs, the gap between the two continuation values. This is Section 1's mechanism inside the quantitative model: because π_d rises in v, how much to use is not decided on the budget alone.

There are two cases in which the third term is zero, and in both the household faces no such trade off: one that abstains, where $v = 0$, and one that already carries the disorder, whose recovery hazard (21) does not depend on how much it uses. For both the condition is the Cobb-Douglas split in closed form,

$$c = \max\left\{F,\ \frac{1-\gamma-\theta}{1-\gamma}\big(x + p_v b\big)\right\}, \qquad v = \max\left\{0,\ \frac{x-c}{p_v}\right\}, \tag{33}$$

whose corner is the abstention decision: a household buys nothing when the Stone-Geary endowment already exceeds what it wants, which is how a large b deters use and is the threshold of Section 1.2.4. The outer maximum on c is the food stamp floor. Where it binds the Cobb-Douglas split no longer describes the allocation: consumption is held at F and the drug takes what the budget leaves, so the floor changes behaviour and not only welfare. Where the third term is present the answer lies between (33) and abstention, since risk can only push towards less, so it is found by a one dimensional search on that interval.

Hours then solve

$$u_c(c)\,\frac{\partial x}{\partial n} \;=\; \omega_\ell (1-n)^{-\sigma_\ell}, \tag{34}$$

and the clawbacks in (17), at a rate of one for food stamps and κ_r for the voucher, make $\partial x/\partial n$ jump *up* at the hours where each transfer runs out, because the clawback stops. That is the welfare trap, and it is the channel through which conditionality changes labour supply rather than only consumption.

2.12 The government

The government pays social security, unemployment insurance, disability, food stamps and vouchers, and supplies a public good $\bar{G}$ that does not enter private consumption and is held fixed across simulations, so that policies are compared at a constant level of it. It finances this with the two taxes in (16) and the estates of the deceased, with the level τ_ℓ clearing the budget in every simulation,

$$\underbrace{\textstyle\sum y^g - \tau_\ell \sum (y^g)^{\xi} + \tau_k \sum ra + \text{estates}}_{\text{revenue}} \;=\; \underbrace{SS + UI + DI + F + Vch + \bar{G}}_{\text{outlay}}. \tag{35}$$

In the stylized model the two policies happen to cost the same and the comparison is unaffected. Here a policy that houses more households spends more, τ_ℓ rises to pay for it, and the higher tax lowers the after tax return to an extra hour: it enters (34) through $\partial x/\partial n$, which is net of the tax schedule in (16), so hours fall and saving with them, for everyone, including the households the programme never reaches, and it runs against Housing First.

2.13 The drug market

Drug production has cost $C(S_v) = \frac{\nu_1}{\nu_2} S_v^{\nu_2}$, where S_v is the aggregate quantity. Supply satisfies

$$p_v = \nu_1 S_v^{\nu_2 - 1}, \qquad S_v = \int v(s)\,\mathrm{d}\Phi(s), \tag{36}$$

with price elasticity of supply $1/(\nu_2 - 1)$ and ν_1 normalised so that the quantity supplied equals the quantity demanded at the benchmark price, which is normalised to one. The supply elasticity, its range and the evidence behind them are set in Section 3.

2.14 The law of motion and newborns

Given the policy functions, the distribution over states evolves by pushing households through their choices and then through the shocks,

$$\Phi_{j+1}(\mathcal{S}') \;=\; \phi(j,\cdot)\int \Gamma\big(\mathcal{S}' \mid s,\, n_j(s), h'_j(s), v_j(s), \tau_j(s)\big)\,\mathrm{d}\Phi_j(s) \qquad \text{for every measurable } \mathcal{S}', \tag{37}$$

together with the newborn inflow

$$\Phi_1(\mathcal{S}') \;=\; \int \big[1-\phi(j,\cdot)\big]\,\mathcal{N}\big(\mathcal{S}' \mid s\big)\,\mathrm{d}\Phi_j(s) \;+\; \int \mathcal{N}\big(\mathcal{S}' \mid s\big)\,\mathrm{d}\Phi_J(s), \tag{38}$$

where $\mathcal{N}$ places a newborn according to (39), so that the mass of newborns equals the mass of the dead exactly and the two equations together conserve the population.

Here $\phi(j,\cdot)$ is evaluated at the household's own $(h'(j,s), d, m)$. The mass that does not survive, plus the mass at $j = J$, is replaced by newborns at $j = 1$ with zero assets, clean, without SMI and without a voucher, and subject to the same unemployment shocks as anyone else, whose productivity is drawn from the unconditional distribution Ω_y of permanent productivity across households scaled by the transmission penalties δ_d for a child whose parent carried the SUD and δ_h for one whose parent was homeless,

$$y_{\text{child}} \;\sim\; \Omega_y \;\text{ scaled by }\; (1-\delta_d)^{\mathbf{1}\{\text{parent had the SUD}\}}\,(1-\delta_h)^{\mathbf{1}\{\text{parent was homeless}\}}. \tag{39}$$

2.15 Competitive equilibrium

A competitive equilibrium is a collection of

- value functions $\big\{\{V_t(j,\cdot)\}_{j=1}^{J}\big\}_{t=0}^{\infty}$;
- policy functions $\big\{\{n_t(j,\cdot),\, h'_t(j,\cdot),\, v_t(j,\cdot),\, \tau_t(j,\cdot),\, a'_t(j,\cdot)\}_{j=1}^{J}\big\}_{t=0}^{\infty}$;
- distributions $\big\{\{\Phi_t(j,\cdot)\}_{j=1}^{J}\big\}_{t=0}^{\infty}$; and
- prices, policies and aggregates $\big\{\tau_{\ell t},\, p_{vt},\, \bar{y}_t,\, y_t^{\text{med}},\, H_t\big\}_{t=0}^{\infty}$,

such that, at every date t,

1. the policy functions solve (30) given its terminal condition, and $\{V_t(j,\cdot)\}_{j=1}^{J}$ are the associated values;
2. $\Phi_{t+1}(j{+}1,\cdot)$ follows from $\Phi_t(j,\cdot)$ by the law of motion (37) together with the newborn rule (39) and (38), and the mass of newborns equals the mass of the deceased;
3. the government budget (35) holds at $\tau_{\ell t}$;
4. the drug market clears at p_{vt}, so (36) holds with aggregate drug demand $S_{vt} = \int v_t(s)\,\mathrm{d}\Phi_t$;
5. the public good is supplied at its assigned level, $G_t = \bar{G}$;
6. $\bar{y}_t$, median gross income y_t^{med} and the homelessness rate H_t are the aggregates implied by Φ_t.

A stationary equilibrium is the special case in which all of these are constant and $\Phi_{t+1} = \Phi_t$, which is what the calibration and the policy comparisons report. The sequence version is stated for completeness, and because the transition exercise in the results needs it.

3 Calibrating the quantitative model

The model has fifty four parameters, some of which vary in value by age. Thirty two are calibrated externally, from data or from the literature, and twenty two are calibrated internally, inside the model. Four further parameters govern drug testing and detection; they are inert in the benchmark, where nothing is conditional, so they are set with the Treatment First experiment rather than here. They are split across the three tables of this section: seventeen carry the key results (Table 5, thirteen internally and four externally calibrated), thirteen are specific to this paper without carrying a key result (Table 6, seven internally and eleven externally), and twenty four are standard (Table 7, seven internally and seventeen externally).

The tables carry the parameter, its description, the moment and source used to fix its value, and the value itself. The text alongside each table adds what a table has no room for: the intuition for what each parameter does, and why its moment identifies it.

3.1 The seventeen parameters that carry the key results

Most of the fifty four are the furniture of any life cycle model. A much smaller set is specific to this paper, and seventeen of those carry its key results. They are set out in Table 5 against the three questions of Section 1.1, with the parameter, its description, and the moment and source used to fix its value, which is given in the last column. They are discussed one at a time below because they are the parameters a reader may want to argue over, and each is taken in the order of the table.

Table 5: The seventeen parameters that carry the key results

Parameter	Description	Set	Moment and source	Value
Who uses drugs, and why				
b	Whether a household uses	Int.	Drug participation, NSDUH 2024	—
λ	SUD hazard per unit used	Int.	Users acquiring a SUD, NSDUH 2024	—
$\bar{\kappa}_{\phi d}$	Mortality with a SUD	Ext.	Mortality ratios, Hjorthøj et al. (2023)	8.00
$\aleph$	Flow value of being alive	Int.	Value of a statistical life, Viscusi and Aldy (2003)	—
ρ_d	Earnings lost to a SUD	Int.	Employment rate of the homeless, HUD 2023	—
ρ_m	Earnings lost to SMI	Int.	Earnings of workers with SMI, NSDUH 2023	—
κ_{ued}	Employment multiplier, SUD	Int.	Unemployment rate given a SUD, NSDUH 2024	—
κ_{uem}	Employment multiplier, SMI	Int.	Employment rate given SMI, NSDUH 2023	—
How housing assistance shapes that decision				
π_V	Voucher lottery probability	Int.	Recipients over eligible, HUD 2023	—
μ_H	Recovery on the street	Int.	Weeks abstinent, Milby et al. (2005)	—
h_1	Service flow of homelessness	Int.	Homelessness rate, HUD 2023	—
$\kappa_{\phi H}(j)$	Mortality when unhoused	Ext.	Mortality ratios by age, Baggett et al. (2013)	4.41 to 1.09
The externality on the child				
ω	Weight on the child	Int.	Parent use rate with a child, NSDUH 2024	—
δ_d	Child's earnings, parent's SUD	Int.	Child SUD ratio, McGovern et al. (2023)	—
δ_h	Child's earnings, homelessness	Ext.	Poverty exposure effects, Chetty et al. (2016)	0.31
The other externalities				
ω_ℓ	Weight on leisure	Int.	Average hours worked, Guvenen et al. (2023)	—
κ_x	Disamenity of homelessness	Ext.	Willingness to pay, Loubière et al. (2020)	0.0008

Who uses drugs, and why.

b, whether a household uses at all. The marginal value of the drug goes as $\theta/(b+v)$, so a larger b makes the first units worth less and a household with little to spend buys nothing. Its value is chosen so the model generates a drug participation rate of 5.13 percent, the NSDUH 2024 figure for adults aged 26 and over.

λ, the hazard per unit used. This is what makes using *more* genuinely riskier rather than merely dearer. Its value is chosen so the model generates the share of users who go on to acquire a SUD, from the same NSDUH 2024 sample. The flow is the moment rather than the stock because prevalence cannot separate onset from recovery: a high prevalence is equally consistent with fast onset and with slow recovery. Recovery itself is not calibrated here at all; it is set externally in Section 3.2 below, which is what leaves this moment free to identify onset.

$\bar{\kappa}_{\phi d}$, mortality with a SUD. This is the parameter the whole paper turns on, and Section 1.1 shows why. Mortality truncates the value of life for a user, and it truncates it hardest for households with the most life left to value. That is what reverses the income gradient and makes the poor the users rather than the rich, which no budget constraint alone delivers. It is calibrated externally, at the midpoint of the standardised mortality ratios for SUD reported in Hjorthøj et al. (2023), which run six to ten, with the age profile taken from CDC WONDER 2023. Setting it externally rather than fitting it is what

frees the elasticity of participation with respect to the drug price to serve as a test of the mechanism rather than being spent pinning a parameter.

ℵ, the flow value of being alive. This is the other half of the mortality mechanism, so it sits beside $\bar{\kappa}_{\phi d}$. $\bar{\kappa}_{\phi d}$ sets how much a SUD raises the chance of dying; ℵ implies what value is being truncated, which is what reverses the income gradient. It needs a level and not only a behavioural target: if the value of life is negative, then households prefer death outright and the mortality cost of using becomes a benefit rather than a deterrent, which can perhaps be used to explain suicide, though that is outside the scope of this paper. Its value is chosen so the model generates a value of a statistical life of $7.5m, from Viscusi and Aldy (2003).

ρ_d and ρ_m, earnings lost while still in work. These belong here rather than among the secondary parameters, because they act on the use decision and not only on who is poor. Earnings enter the budget, so a household that loses income to a disorder can afford less of everything including the drug; and earnings enter the continuation value, so losing them lowers what a household has to lose from dying, a similar channel to the one ℵ works through. They also do a job nothing else can: they let an *employed* household be poor enough to be priced out of housing. An effect on the employment margin alone cannot deliver that, since it makes every homeless household jobless and about half are not. One subtlety is worth stating, because it cuts against Section 1.1. There the *proportional* income penalty was shown to matter little for whether a household uses, and that remains true holding the Stone-Geary endowment fixed in ratio terms. But b is a level, not a share, so conditional on b a larger penalty lowers x against a fixed subsistence endowment, and the corner in (33) binds for more households. So raising ρ_d does reduce use at the bottom; what Section 1.1 rules out is the penalty *generating* the income gradient, not its having any effect at all. ρ_d's value is chosen so the model generates the employment rate of the homeless, about half, from HUD's 2023 point-in-time count and TEDS-A 2022; ρ_m's so it generates the earnings of workers with SMI relative to average, from NSDUH 2023.

κ_{ued} and κ_{uem}, the employment multipliers. These belong here for the same reason as the wage penalties: work is harder to find and easier to lose with a SUD or with mental illness, and losing work lowers both the budget and the continuation value. They are the margin that turns a health shock into an income shock, and without them a disorder would cost a household nothing beyond the drug itself. Their values are chosen so the model generates the unemployment rate given a SUD, 16.25 percent in NSDUH 2024, and the employment rate given SMI, about 30 percent.

How housing assistance shapes that decision.

π_V, the chance of drawing a voucher. This is the policy dial, and it does three things at once, which is why it cannot be read as generosity alone. It lowers homelessness mechanically, since a household with a voucher can afford a dwelling it otherwise could not. It raises the value of being alive, because a household with a voucher has a better life ahead of it and so more to lose from the mortality that use brings, and that pushes against using. And running in the opposite direction, it insures the household against the consequences of using: a voucher that arrives whatever a household does removes part of the penalty for ending up unable to pay rent, and that is the moral hazard the paper is about. So the same instrument that gives a household more to protect also weakens its reason to protect it, and which of the two dominates is quantitative. Its value is chosen so the model generates the 26.6 percent ratio of voucher recipients to the income eligible population in HUD's Picture of Subsidized Households, 2023.

μ_H, how much harder recovery is on the street. Housing enters recovery here and not only the budget, and this is the parameter that carries it. It matters more than the housed recovery rate because it is a *gradient*: it is how much a household gains in recovery terms from being housed, so it is what a voucher buys beyond shelter, and that makes it the counterweight to the moral hazard in π_V. If μ_H is large, assistance buys recovery and the insurance effect is outweighed; if it is near one, assistance is a transfer and the moral hazard stands unopposed. Its value is chosen so the model generates the increase in recovery that housing produced in Milby et al. (2005), a randomised trial on cocaine dependent homeless adults: 6.60 weeks abstinent in the housed arm against 4.48 unhoused, so housing raises recovery by 47.3 percent.

h_1, **the service flow of homelessness.** It belongs among the key parameters because it sets how bad the outside option is, and the value of being alive is measured against that outside option: a household with little to lose from dying is a household for which the mortality cost of using is small. It also decides the welfare comparison directly. If Housing First houses more households and the cost of homelessness is high, Housing First can deliver higher welfare than Treatment First even once the externalities on the child are counted against it. Its value is chosen so the model generates the homelessness rate, 0.399 percent of households in HUD's 2023 point-in-time count.

$\kappa_{\phi H}(j)$, **mortality when unhoused.** This is how much of the cost of dying a voucher removes, which is why it belongs here and why it runs against Treatment First: a policy that houses more people saves lives directly, whatever it does to disorders, although in general equilibrium it can backfire by raising the SUD rate. Baggett et al. (2013) give race adjusted mortality ratios for homeless adults of about 9.1 at ages 25 to 44, falling to 1.1 at 65 to 84. Those are totals for a population most of which also carries a SUD, so it is set externally age by age net of that composition, which is why it has to be derived jointly with $\bar{\kappa}_{\phi d}$ rather than read off directly, as set out at (25).

The externality on the child. These three are the externalities that a parent's SUD and a parent's homelessness impose on its child. The first runs in favour of Treatment First and the second in favour of Housing First, which is why they are taken together.

ω, **the parent's weight on its child.** A parent values its child's lifetime utility at ω times its own, and ω is small. What that means for policy is the point: the two penalties below cost the child in full, the parent counts only ω of that when deciding whether to use, and the remaining $1-\omega$ is a cost nobody in the model weighs in their decisions. Identifying it needs care, because *every* household here has a child, so there is no parent against non parent comparison to run inside the model. The moment is a counterfactual instead. In NSDUH 2024 past month use is 15.5 percent lower among parents than among non parents, with parents reweighted to the non parents' joint distribution of age, income, sex and marital status so the gap is behaviour rather than composition. Since ω is the only reason a household in the model uses less than it otherwise would, its value is chosen so that switching the child term off at the calibrated parameters raises drug use by that same 15.5 percent.

δ_d, **what the child loses to a parent's SUD.** This one argues for Treatment First. McGovern et al. (2023) find that children of a parent with a SUD are two to four times likelier to acquire one themselves. The model does not carry an inherited disorder; it carries the loss of earnings potential that goes with it. No paper estimates the effect of a parent's SUD on a child's lifetime earnings directly, which is why the penalty is calibrated to the disorder ratio rather than lifted from an estimate. What the literature supplies is the two halves of the chain: Felitti et al. (1998) link household dysfunction, of which parental substance use is a component, to adult health and behavioural outcomes, and Chetty et al. (2016) measure what childhood environment does to adult earnings. δ_d is the reduced form of the two. δ_d is calibrated internally, its value chosen so the model reproduces McGovern's ratio of child disorder rates between parents with and without a disorder. A policy that cures disorders therefore earns a return on the next generation that a policy which only houses people does not.

δ_h, **what the child loses to growing up homeless.** This one argues for Housing First, and it comes from the same literature read in the other direction. Chetty et al. (2016) find that moving a child to a lower poverty neighbourhood before age thirteen raises adult earnings by 31 percent; δ_h is calibrated externally to that figure, taking the earnings gain from leaving poverty as the mirror of the loss from growing up without stable housing. So the two penalties pull in opposite directions, and which return is larger is exactly the quantitative question the paper asks.

The other externalities. These two act on households who are not the children, and neither runs through the drug decision directly.

ω_ℓ, **the weight on leisure.** This sets how much labour supply responds to the tax, and to the programme's own design. A policy that houses more households is financed by a higher τ_ℓ, which lowers the after tax return to an hour worked, so hours fall for households the programme never reaches. The design

pulls the same way for those it does reach: the voucher is clawed back at rate κ_r as income rises, so a recipient who works more keeps less, and works less as a result. And the channel runs back the other way, which is what makes it an externality rather than only a cost: a household that uses is likelier to be out of work and likelier to need assistance, so its choice raises the tax everybody else pays. Its value is chosen so the model generates average hours worked, from Guvenen et al. (2023).

κ_x**, the disamenity of homelessness.** It is the only term through which one household's housing status enters the utility of another household who is not its own child, which is what makes the homelessness rate a genuine fixed point of the equilibrium rather than accounting. It is calibrated externally from the contingent valuation in Loubière et al. (2020): €57 a year among the 42 percent of respondents willing to pay anything, which is 0.20 percent of average annual earnings, against €23 and 0.08 percent averaged over all respondents. The higher figure is the baseline. That is the conservative choice here rather than the generous one, because the disamenity adds to whichever policy houses more people: setting it high loads the comparison in favour of Housing First, so a ranking that still favours Treatment First survives the assumption least kind to it.

3.2 Other parameters specific to this paper

Thirteen further parameters are specific to this paper, and none of them carries a key result. Table 6 collects them and marks which are internally and which externally calibrated. The table is broken into the same six groups the discussion follows, and each parameter is taken in turn below.

Table 6: Parameters specific to this paper

Parameter	Description	Set	Moment or source	Value
The employment margin for homelessness				
κ_{ueH}	Employment multiplier, homeless	Ext.	Desmond and Gershenson (2016)	0.80
Recovery and treatment				
$\bar{\pi}_r$	Recovery when housed	Ext.	Ten year remission rate, Vaillant (1995)	0.025
g	Treatment effect on recovery	Ext.	Odds ratio, Milby (2005)	1.35
$\bar{\tau}^u$	Utility cost of treatment	Int.	Treatment take up, NSDUH 2023	—
The mental illness chain				
p_m	Prevalence of SMI among adults	Ext.	Share with serious mental illness, NSDUH 2023	5.5%
λ_m	Recovery rate from SMI	Int.	SMI share of homeless, Gutwinski et al. (2021)	—
ι	SUD multiplier on the onset of SMI	Ext.	Danish registers, Nielsen et al. (2017)	2.00
μ_m	Recovery multiplier, SMI	Ext.	Dual diagnosis relapse, Bradizza et al. (2006)	0.82
$\kappa_{\phi m}$	Mortality multiplier, SMI	Ext.	Olfson et al. (2015)	2.50
ζ_{DI}	Disability replacement rate, SMI	Ext.	SSDI and SSI rates (SSA 2023)	0.25
The drug market, and the value of life				
ν_1	Drug supply shifter	Ext.	Normalisation, so $p_v = 1$ in the benchmark	norm.
p_v	Price of drugs	Ext.	Clears the drug market in the benchmark	1.00
ν_2	Drug supply elasticity	Ext.	Elasticity 2; Hansen et al. (2017), Chaloupka (1991)	1.50

Earnings, conditional on working.

κ_{ued}**, the employment multiplier for a SUD.** Work is harder to find and easier to lose with a disorder, and this is what converts a health shock into an income shock. Its value is chosen so the model generates the unemployment rate among those with a SUD, 16.25 percent in NSDUH 2024.

κ_{uem}**, the employment multiplier for SMI.** The same construction for mental illness, its value chosen so the model generates the employment rate given SMI, about 30 percent, which is the midpoint of the range reported for high disability diagnoses.

κ_{ueH}**, the employment multiplier for being unhoused.** Being on the street makes work harder to hold, independently of why one is there. It is set externally, because separating it from the disorder in the data would need the kind of randomised variation in housing that does not exist at this margin.

Recovery, and the treatment decision.

$\bar{\pi}_r$, **recovery when housed.** The baseline hazard that μ_H scales. It sets the *level* of recovery rather than its gradient in housing, which is why it is not among the key parameters, and it is set externally from the ten year remission rate in Vaillant (1995), bracketed by Fleury et al. (2016).

g, **the treatment effect on recovery.** Set externally from the odds ratio for contingency management, and reported across a range rather than at a point because it is the one externally set value carrying a headline result.

$\bar{\tau}^u$, **the utility cost of treatment.** Treatment raises recovery, so without a cost every household with a disorder would take it. This is what stops that, and it is the parameter that makes take up a decision rather than an assumption. Its value is chosen so the model generates treatment take up among those with a SUD, 24 percent in NSDUH 2023.

The mental illness chain.

p_m, **the prevalence of SMI.** Set externally at 5.5 percent of adults from NSDUH 2023, defined by substantial functional impairment rather than any mental illness, which is roughly a quarter of adults. A state that halves productivity cannot sensibly cover a quarter of the population.

λ_m, **the rate of leaving SMI.** This one needs care. The onset profile is backed out of the observed prevalence profile, so prevalence is reproduced by construction at any λ_m and cannot identify it. What identifies it is *persistence*: only a sufficiently sticky state lets the SMI population accumulate in homelessness at four and a half times its share of adults. Its value is chosen so the model generates the SMI share of the homeless, 25 percent in Gutwinski et al. (2021).

ι, **the SUD multiplier on SMI onset.** A substance use disorder raises the hazard of mental illness. This is the channel, and it runs the direction the evidence supports: in the Danish registers any SUD predicts conversion to schizophrenia with a hazard ratio of 1.34, opioid use disorder 2.74 (Nielsen et al., 2017). It is a further private cost of a disorder and a further externality, since SMI raises disability receipt and mortality.

μ_m, **the SMI recovery multiplier.** Mental illness slows recovery from a disorder. Set externally from dual diagnosis relapse of 39.8 against 26.4 percent, and flagged as one of the three values carrying no real discipline.

$\kappa_{\phi m}$, **mortality with SMI.** Set externally at 2.50, and flat in age, because Olfson et al. (2015) report standardised mortality ratios for schizophrenia of 3.6, 3.4 and 4.0 across the working ages. Flat, if anything rising, so the absence of an age argument is a finding.

ζ_{DI}, **the disability replacement rate.** What a household out of work with SMI receives. Set externally from SSDI and SSI replacement rates, and it is the transfer that İmrohoroğlu and Zhao (2022) identify as the effective policy for the chronic group, which is why it is in the model at all.

The drug market, and the value of life.

ν_2, **the supply elasticity.** Sets the price elasticity of supply at 2, the middle of a range from about 0.7 to about 4. There is no direct estimate for an illicit drug, so it is bracketed by the closest legal markets: the Washington State marijuana reform implies roughly 0.79, the cigarette and gasoline literatures nearer 3.6.

ν_1, **the supply shifter.** A pure normalisation, chosen so that supply equals demand at $p_v = 1$ in the benchmark. It normalises units, not behaviour.

p_v, **the drug price.** One at the benchmark by that normalisation, and free in every counterfactual. That is what lets a policy which changes how much is used move the price, and so feed back into every other household's decision.

3.3 Standard parameters

The rest are conventional, recorded so the calibration can be reproduced rather than because anything turns on them. Table 7 carries each with its source and vintage, in this order.

Table 7: Standard parameters

Parameter	Description	Set	Moment or source	Value
β	Discount factor	Int.	Wealth over output, BEA	—
γ	CD weight on housing	Int.	Housing cost over income, SCF 2022	—
θ	CD weight on drugs	Int.	Drug spending given a SUD, NSDUH 2024	—
$\underline{h}$	Smallest rental size	Int.	Housing cost over half income, SCF 2022	—
ς	Geometric spread of rentals	Int.	Housing cost over 30 pct, SCF 2022	—
$\underline{c}$	Food stamp consumption floor	Int.	Food stamps over labour income, USDA	—
$\bar{\pi}^{eu}$	Job separation rate	Int.	Unemployment rate, BLS CPS 2014–24	—
σ	Risk aversion	Ext.	Attanasio and Weber (2010)	2.00
σ_ℓ	Curvature on leisure	Ext.	Frisch elasticity of 0.5, Chetty (2012)	3.00
$\varepsilon(j)$	Age profile of efficiency	Ext.	Quadratic in age, PSID and CPS hump	profile
$\bar{\phi}(j)$	Baseline survival by age	Ext.	SSA period life table, ages 26 to 85	profile
ρ_y	Persistence of log productivity	Ext.	Storesletten, Telmer and Yaron (2004)	0.90
σ_y	Dispersion of log productivity	Ext.	Storesletten, Telmer and Yaron (2004)	0.20
j_E	Maximum experimentation age	Ext.	Reported with and without it	55
r	Risk free rate	Ext.	Kaplan, Mitman and Violante (2020)	0.03
ξ	Tax progressivity	Ext.	Heathcote et al. (2017)	0.85
τ_k	Capital income tax	Ext.	US long run rate on capital gains and	0.15
τ_ℓ	Tax level	Ext.	Clears the government budget	in eqm.
G	Other public spending	Ext.	Residual at benchmark, then held fixed	in eqm.
ψ	UI replacement rate	Ext.	DOL statutory replacement rates	0.50
$\bar{y}$	Cap on the insurance earnings base	Ext.	DOL state benefit maxima	1.00
κ_V	Voucher eligibility cutoff	Ext.	HUD rule	0.50
κ_r	Tenant contribution rate	Ext.	HUD rule	0.30
$\bar{\pi}^{ue}$	Job finding rate, annual	Ext.	Krueger, Cramer and Cho (2014)	0.55

β, σ, σ_ℓ. Discount factor, value chosen to match wealth over output; risk aversion and the curvature on leisure, both standard.

γ and θ. The Cobb-Douglas weights on housing and the drug, each chosen so the model matches its own expenditure share.

$\underline{h}$ and ς. The rental ladder, which is geometric, so these two numbers give the whole grid and no size is set by hand. They are matched to the shares of renters spending more than half and more than thirty percent of income on housing, which separates a level from a spread. A uniform grid is a restriction on ς, not a neutral choice, and fits those two shares worse.

$\underline{c}$. The food stamp floor, chosen so the model matches programme spending.

$\bar{\pi}^{eu}$ and $\bar{\pi}^{ue}$. Baseline separation and job finding rates.

$\varepsilon(j)$, $\bar{\phi}(j)$, ρ_y, σ_y, j_E. Age profiles of earnings and survival, the persistence and dispersion of permanent productivity, and the age at which use can begin.

r, ξ, τ_k, τ_ℓ, G. The interest rate and the tax system, with τ_ℓ not set at all but whatever clears the budget, and G held fixed across simulations.

ψ, $\bar{y}$, κ_V, κ_r. The replacement rate and cap for unemployment insurance and the eligibility and contribution rates for the voucher, all set by statute.

3.4 How the calibration is done, and what it cannot do

The first stage matches nineteen parameters to nineteen moments of the stationary equilibrium; the second matches short run responses measured in experiments, which no comparison of steady states identifies, and carries μ_H and ω. Table 9 reports the data moment and the model moment side by side for every calibrated parameter, so the fit can be inspected rather than asserted, and each is solved to a distance of 10^{-3}, or 10^{-4} for the three bisected below.

Two things the calibration does not do are worth stating. Three externally calibrated values carry no real discipline, and they are named rather than buried: the treatment effect g, the SMI recovery multiplier μ_m and the drug supply curvature ν_2. Of these only g carries a headline result, and it is reported across a range. And the drug price participation elasticity is deliberately not a target: with $\bar{\kappa}_{\phi d}$ assigned from mortality data, the elasticity is computed after the fact, between the benchmark and a re solved economy with the supply curve shifted, and compared with the range estimated in the literature. It tests the mechanism instead of constraining it, which is only possible because the drug price is endogenous; ν_1 normalises the units so that supply meets demand at $p_v = 1$ in the benchmark, and p_v moves freely in every counterfactual.

Three moments that are cliffs, not slopes. Most parameters move their moment smoothly, so a step in one direction tells you where to go next. Three do not, and treating them as though they did is how a calibration silently fails. Three moments are step like and are bisected rather than stepped, for the reason Section 1.3 gives: participation is a near threshold function of b, homelessness moves from 0.12 to 3.29 percent for a 35 percent change in h_1, and wealth over output is 5.46 at $\beta = 0.995$ and 0.38 at $\beta = 0.85$. All three are bisected at the head of each sweep, in partial equilibrium, before anything else moves. The parameter to moment map is close to triangular and the sign of each parameter's effect on each moment is declared for every pair, so a mistaken sign shows up as a moment walking away from its target rather than as a silent failure.

The disamenity of homelessness. The disamenity of aggregate homelessness is priced rather than assumed, and the evidence for it is thin enough to be worth stating exactly. Loubière et al. (2020) ask adults in eight European countries what they would pay in annual taxes to end homelessness through Housing First: the mean over all respondents is €23 a year, and €57 among the 42 percent willing to pay anything, which is 0.08 and 0.20 percent of average annual earnings. κ_x is set so that households in the model would pay the first of those to move H to zero, and the second is the top of the range the results are reported across.

The disamenity is on the headcount alone. There is no extra weight on the homeless who happen to carry a SUD, and that is deliberate: the survey evidence asks what people would pay to end *homelessness*, so that is the object priced, and loading the SUD into the same term would price something the evidence was not about.

Two things follow. The term is small, so it cannot by itself decide a ranking; what it can do is add to whichever policy houses more people, which is why reporting the range matters more than the point estimate. And it is the only place where a household's utility depends on an aggregate, which is what makes H a genuine fixed point of the equilibrium rather than accounting.

Anchors. The discount factor targets aggregate wealth at three times output, with output measured as all factor income and transfers excluded, which is the standard US private capital to output ratio rather than the SCF's net worth to household income ratio of about 7.3, whose numerator is mostly housing equity a model without homeownership cannot hold. As in Section 1.3, $\aleph$ is set to a value of a statistical life rather than to a behavioural moment, because a behavioural target pins $\aleph$'s effect on *who* uses without pinning its level, and at the wrong level households prefer death and the mortality cost of drug use becomes a benefit.

Three targets that were wrong and are now right. Substance use disorder among the homeless is 21.7 percent, the *drug* use disorder figure from Gutwinski et al.'s pooling of thirty nine studies, not the 36.7 percent that is alcohol. The income of drug users is split: 95.1 percent of average for users and 68.7 percent for a moderate or severe SUD, ages 26 and over, with the SUD ratio targeted and the use ratio kept for validation, which makes the economics explicit, since use is close to income neutral and it is progression to the SUD that concentrates at the bottom. The housing burden is targeted over all households rather than over renters only.

Second stage: responses, not steady states. These are partial equilibrium cohort experiments. A trial of two hundred people does not move the tax rate or the drug price, so prices are held at benchmark; the household problem is solved again, because each arm changes the rules a forward looking household faces. Milby et al. (2005) randomised 196 cocaine dependent homeless adults into three arms on identical day treatment, and the targeted moment is the ratio of the unconditional voucher arm to no assistance, $6.60/4.48 = 1.473$ weeks abstinent, which is what identifies μ_H. The conditional against unconditional comparison is deliberately *not* targeted, because the source reports it as insignificant in the raw randomised comparison; calibrating to it would repeat the error this revision is correcting elsewhere. The drug price participation elasticity, -0.63, is no longer targeted. It is reported below as a moment the model is not fitted to, which is the stronger use of it: with $\bar{\kappa}_{\phi d}$ assigned from mortality data, the elasticity tests whether the mechanism delivers the observed price response rather than being spent on pinning a parameter. It remains a moment only an endogenous drug price can produce, since a fixed price has no elasticity to match.

What the child parameters are matched to. One of the two is calibrated. The weight on the child ω carries the behavioural moment; the transmission penalty δ_d is assigned, for the reasons given at the head of this section. That also removes a joint identification problem, since the two used to have to be fitted together: altruistic parents use less, so the share of children born to a parent with the SUD falls, so the transmission ratio at a given δ_d fell with ω. With δ_d assigned, ω is identified from its own moment alone.

Parameter	Moment	Source
ω	Drug use falls 15.5 percent with a child present	NSDUH 2024, ages 26+, parents reweighted to the non parents' joint age, income, sex and marital distribution
δ_d	Set, not calibrated	McGovern et al. (2023), midpoint of the epidemiological range of two to four, reported with a sensitivity band across it
δ_h	Set, not calibrated	Chetty et al. (2016)

Since every household in the quantitative model has a child, there is no parent and non parent comparison to run inside it. The condition is the counterfactual instead: ω is the entire reason a household uses less than it otherwise would, so switching altruism off at the calibrated parameters must raise drug use by 18.4 percent. Run through the stylized model that condition gives $\omega = 0.014$, which is worth stating because it is small: the ranking in Section 1.6.2 reverses back to Housing First only near $\omega = 0.43$, so the calibrated weight sits a factor of thirty inside the region where the externality still decides the comparison, and it is consistent with Altonji, Hayashi and Kotlikoff (1997), who reject altruism operating at the margin at all. The measured gap is an upper bound, since users also lose custody of children, so the ranking is reported across zero to that value.

Tables 8 and 9 list every parameter of the model: those that are assigned, whether by normalisation, from the literature, directly from data, in equilibrium or as a reported sensitivity, and those that are calibrated inside the model, each against the moment it is assigned to. The full derivation behind each value, which used to sit inside the tables, is in `writing/old/`; what is printed here is the parameter, what it means, where the number comes from and the number itself.

Table 8: Externally calibrated parameters of the quantitative model

Parameter	Description	Source or rationale	Value
Preferences and demographics			
σ	Risk aversion	Attanasio and Weber (2010); the standard life cycle value	2.00
σ_ℓ	Curvature on leisure	Frisch elasticity of 0.5, Chetty (2012)	3.00
$\varepsilon(j)$	Deterministic age profile of efficiency, varies with age	Quadratic in age, fitted to the PSID and CPS hump	profile
$\bar{\phi}(j)$	Baseline survival, varies with age	SSA period life table, ages 26 to 85, the sexes combined by survivor weight	profile
ρ_y, σ_y	Persistence and dispersion of log productivity	Storesletten, Telmer and Yaron (2004); the stochastic process is the robustness direction, the baseline holds y fixed	0.90, 0.20
j_E	Maximum experimentation age	Reported with and without it	55
Prices, taxes and the government			
r	Risk free rate	Kaplan, Mitman and Violante (2020)	0.03
ξ	Tax progressivity	Heathcote et al. (2017)	0.85
τ_k	Capital income tax	US long run rate on capital gains and qualified dividends	0.15
τ_ℓ	Tax level	Clears the government budget	in eqm.
G	Other public spending	Residual at the benchmark, then held fixed so that policy is financed by τ_ℓ	in eqm.
κ_x	Disamenity of the homelessness rate, headcount only	Loubière et al. (2020): €23 a year on average, 0.08 percent of average earnings, to end homelessness; the €57 among those willing to pay anything is the top of the reported range	priced
Income support and the voucher			
ψ	Unemployment insurance replacement rate	DOL statutory replacement rates	0.50
$\bar{y}$	Cap on the earnings base for insurance, share of average income	DOL state benefit maxima	1.00
κ_V	Voucher eligibility threshold, share of median income	HUD rule	0.50
κ_r	Tenant contribution rate	HUD rule	0.30
p_{test}	Probability a voucher holder is tested	Benchmark value, swept downward	1.00

Parameter	Description	Source or rationale	Value
c_{test}	Cost per voucher holder tested	A urine panel at about $40, quarterly	0.0023
p_{det}	Detection probability given a test	Benchmark value, swept downward	1.00
p_{fp}	False positive rate under Treatment First	Benchmark value, swept upward	0.00
Employment, the SUD and SMI			
$\bar{\pi}^{ue}$	Job finding rate, annual	Krueger, Cramer and Cho (2014)	0.55
κ_{ueH}	Employment multiplier, homeless	Desmond and Gershenson (2016)	0.80
$\kappa_{\phi H}(j)$	Mortality multiplier, homeless, varies with age	Baggett et al. (2013); backed out age by age from (25)	4.41 at 26 to 1.09 at 85
$\kappa_{\phi m}$	Mortality multiplier, SMI, flat in age	Olfson et al. (2015)	2.50
$\bar{\kappa}_{\phi d}$	Mortality multiplier, SUD, the LEVEL at its peak age; it generates the age profile $\kappa_{\phi d}(j)$ through (25)	Midpoint of the all cause standardised mortality ratios reported for opioid use disorder, which run 6 to 10; the age shape is from CDC WONDER drug poisonings	8.00
$\bar{\pi}_r$	Baseline recovery hazard when housed, annual	Vaillant (1995); bracketed by Fleury et al. (2016)	0.025
μ_m	Recovery multiplier, SMI	Dual diagnosis relapse of 39.8 against 26.4 percent	0.82
g	Treatment effect on recovery	Odds ratio of 1.22 for contingency management	1.35
p_m	Prevalence of SMI among adults	NSDUH 2023, defined by substantial functional impairment	5.5%
ζ_{DI}	Disability replacement rate, non employed with SMI	SSDI and SSI replacement for this group	0.25
ι	SUD multiplier on the onset of SMI	Danish registers, hazard ratios of 1.34 to 2.74	2.00
δ_d	Transmission penalty, parental SUD	McGovern et al. (2023) give a range of 2 to 4, not a point, and δ_h is already assigned from the same literature	3.00
Transmission to the child			
δ_h	Transmission penalty, parental homelessness	Chetty et al. (2016)	0.31
The drug market			
p_v	Price of drugs	Clears the drug market, and is one at the benchmark	1.00
ν_1	Drug supply shifter	Normalisation, so that $p_v = 1$ at the benchmark quantity	norm.
ν_2	Drug supply curvature	Elasticity $1/(\nu_2 - 1) = 2$; Hansen, Miller and Weber (2018) give 0.79 and the cigarette and gasoline literatures about 3.6	1.50

Table 9: Internally calibrated parameters of the quantitative model

Parameter	Description	Target moment	Source	Data	Model	Value
The three step like moments, bisected						
b	Stone-Geary endowment, constant across every state and choice	Drug participation rate	NSDUH 2024, ages 26+	5.13%	—	—
h_1	Service flow of homelessness	Homelessness rate	HUD point in time 2023, households	0.399%	—	—
β	Discount factor	Wealth over output	BEA fixed assets against GDP	3.0	—	—
Anchors						
$\aleph$	Flow value of being alive	Model implied value of a statistical life	Viscusi and Aldy (2003)	$7.5m	—	—
Preferences and the housing ladder						
ω_ℓ	Weight on leisure	Average hours worked	Guvenen et al. (2023)	40%	—	—
γ	Cobb-Douglas weight on housing	Aggregate housing cost over income	SCF 2022	21.2%	—	—
θ	Cobb-Douglas weight on drugs	Drug spending over income given a SUD	NSDUH and ADAM	44%	—	—
$\underline{h}$	Smallest rental size	Share with housing costs above half of income	SCF 2022	17.49%	—	—
ς	Geometric spread of the rental sizes	Share with housing costs above 30 percent of income	SCF 2022	40.62%	—	—
Income support						
$\underline{c}$	Food stamp consumption floor	Food stamps over labour income	USDA, SNAP outlays	0.42%	—	—
π_V	Voucher lottery probability	Recipients over the income eligible	HUD Picture of Subsidized Households	26.6%	—	—
Employment margins						
$\bar{\pi}^{eu}$	Job separation rate	Unemployment rate	BLS CPS, series UNRATE via FRED, 2014 to 2024	4.85%	—	—
κ_{ued}	Employment multiplier, SUD	Unemployment rate given a SUD	NSDUH 2024, ages 26+	16.25%	—	—
κ_{uem}	Employment multiplier, SMI	Unemployment rate given SMI	NSDUH, by Bayes	7.1%	—	—
The wage penalties						

Parameter	Description	Target moment	Source	Data	Model	Value
ρ_m	Productivity penalty of SMI, conditional on working	Earnings of workers with SMI relative to average	to be pinned in the data appendix	—	—	—
ρ_d	Productivity penalty of a SUD, conditional on working	Share of the homeless who are employed	to be pinned in the data appendix	about half	—	—
Treatment, the SUD hazard and SMI						
$\bar{\tau}^u$	Utility cost of treatment	Treatment take up given a SUD	NSDUH 2023	24%	—	—
λ	SUD hazard per unit of use	SUD rate	NSDUH 2024, ages 26+	3.68%	—	—
λ_m	Recovery rate from SMI, annual, flat in age; with the observed prevalence profile it implies the age varying onset $\text{onset}_m(j)$ of (22)	SMI share of the homeless	Gutwinski et al. (2021)	25%	—	—
The second stage, matched to experimental responses						
μ_H	Recovery multiplier, homeless	Housed against unhoused abstinence	Milby et al. (2005)	1.473	—	—
ω	Weight on the child	Fall in drug use with a child present	NSDUH 2024, reweighted	15.5%	—	—

3.5 Validation, and what is left undisciplined

None of the moments below is fitted. Each is produced endogenously by a model with sixty single ages, and each is a place the model can be caught.

Moment	Data	Source
Homelessness rate by age, 25 to 34 through 65+	0.316 / 0.338 / 0.291 / 0.250 / 0.077 percent	HUD PIT 2024
Shares of the homeless across those bands	26.0 / 27.2 / 21.0 / 18.4 / 7.5 percent	HUD PIT 2024
Returns to homelessness within 24 months	17.4 percent	HUD SPM 2024
Drug poisonings as a share of all deaths, 25 to 44	about 20 percent	CDC WONDER
Bottom half's share of net worth	2.52 percent	Fed DFA
Housing costs above 30 and above 50 percent of income	40.6 and 17.5 percent	SCF 2022
Age weighted standardised mortality ratio, SUD	2.5 to 9.8	Degenhardt et al.
Age weighted standardised mortality ratio, homeless	9.1 / 4.5 / 1.1 by band	Baggett et al.
Tenant contribution rate	31.2 percent	HUD
Voucher holders with wages as their main income	28 percent	HUD
Income with a moderate or severe SUD, over average	71.1 percent	NSDUH 2024
Labour force participation, SUD against clean	65.8 against 61.5 percent	NSDUH 2024
Homeless at treatment admission	14.5 percent	TEDS-A 2022
Co-occurring psychiatric problem at admission	43.1 percent	TEDS-A 2022
Working at all, among the homeless in treatment	6.6 percent	TEDS-A 2022
Drug use disorder share of the homeless	21.7 percent	Gutwinski et al. (2021)
Drug price participation elasticity	-0.63	Saffer and Chaloupka (1999)

Three parameters remain undisciplined, and they are flagged rather than hidden: the treatment effect on recovery g, the recovery multiplier for SMI μ_m, and the drug supply curvature ν_2. Of these only g carries a headline result, and it is reported as a range rather than as a number.

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